



Subject Code	CSC2332
Subject Name	SOFTWARE PROJECT MANAGEMENT
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Assignment number &title	Assignment 1 – 20% Case Study Analysis: Applying Project Management Principles Citizen Registration and Big Data Analytics Platform (CRBDAP) Federal Republic of Somalia
I declare that all material in this assignment is my own work except where there is clear acknowledgement or reference to the work of others and I have complied and agreed to the University statement on Plagiarism and Academic Integrity. Signature: _____ Date: 22/5/2026	

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Question 1: Project Management Fundamentals

1.1 Definitions

Project: A project is a temporary activity with a clear start and end date. It is done to create something new, such as a product, service, or result. For Somalia, the CRBDAP project is a government project that aims to build a national system to register all citizens and analyze their data. Once the system is finished and handed over to the government, the project ends.

Project Management: Project Management means using knowledge, skills, and tools to plan and complete a project successfully. It helps the team finish work on time, within budget, and with good quality. For CRBDAP in Somalia, project management is very important because the country has limited resources, weak internet in some areas, and a complex political environment. Good project management helps the team deal with these challenges and deliver the system as planned.

1.2 Two Important Roles of a Project Manager

Role 1 – Planning the Project Scope: The Project Manager must clearly define what the project will do and what it will not do. For CRBDAP in Somalia, this means deciding which government departments, such as NIDA, the Ministry of Health, and the Ministry of Education, will be connected to the system. If the scope is not clear, new requirements will keep coming in, and the project will never finish on time. This problem is called scope creep. By making a clear scope plan from the beginning, the Project Manager protects the team from unnecessary work and keeps the project on track.

Role 2 – Managing Risks: Somalia has many unique challenges, including poor internet in rural areas, political issues in some regions, and limited ICT professionals. The Project Manager must find these risks early, write them in a risk register, and prepare solutions. For example, if internet is not available in Baidoa, the team can prepare an offline version of the registration system. Managing risks early prevents big problems later and helps the project succeed.

Question 2: Project Management Context and Processes

2.1 Importance of a Systems View of Project Management

A Systems View means looking at the whole project as one connected system, not just separate tasks. Everything in the project is related to everything else. If one part changes, it affects other parts.

For the CRBDAP project in Somalia, this view is important for these reasons:

- The system must connect many government departments such as health, education, and civil registration. If one department changes its data format, it affects all other departments. The systems view helps the Project Manager see these connections and plan for them.
- Somalia has different regional governments such as Puntland and Jubbaland. A decision made in Mogadishu can affect work in other regions. The systems view helps the PM involve all regions early to avoid conflicts later.
- Some problems, like community distrust of data collection, can slow down the whole project. The systems view helps the PM see these problems early and add community awareness activities to the plan.

In short, the systems view helps the Project Manager understand the big picture and avoid managing the project in pieces.

2.2 How Organizational Structure and Communication Affect the Project

Organizational Structure: In Somalia, most government ministries work separately and do not share information easily. This is called a functional structure. In this type of structure, the Project Manager has little authority over resources from other departments. This makes it hard to get cooperation from the Ministry of Health or the Ministry of Education when needed. A matrix or projectized structure is better for CRBDAP because it gives the Project Manager more control and makes cooperation between departments easier.

Communication: Poor communication is one of the biggest risks in this project. Somalia has different languages and dialects, and team members are spread across the country. If the development team in Mogadishu does not communicate well with field staff in Baidoa or Garowe, the system may be built without knowing important local needs, such as offline registration for areas with no internet. The Project Manager must create a Communication Management Plan that includes regular meetings, progress reports, and the use of simple tools like WhatsApp groups or Microsoft Teams, which are commonly used in Somalia.

Question 3: Project Integration and Scope Management

3.1 Three Project Integration Management Processes

Process 1 - developing a project charter: The project charter is an official document that grants the project manager the legal right to begin the project and allocate resources. For the CRBDAP project located in Somalia, the project charter could originate from either the prime minister's office or the Ministry of Interior. The project charter provides a description of the project; specifically, its objectives, budget, timeline, and other key players involved with the project (such as NIDA), as well as funding contributors (such as the World Bank and UNHCR). Absent this document, the project lacks formal legitimacy, which may result in team members not taking the project seriously. In addition, within Somalia's present governance framework, establishing credibility and obtaining international partner commitment requires the development of a project charter.

Process 2 - developing a project management plan: A project management plan is the primary document developed by the project team that outlines how the project will be conducted. This document contains elements related to scope, scheduling, cost, risk, and communications. For the CRBDAP project, the Project Management Plan must contain a phase-in approach for the implementation of the registration system beginning with Mogadishu and extending outward to regional capital cities (Garowe, Kismayo, Hargeisa). Additionally, it must outline contingency planning strategies for potential issues (e.g., internet access failures, security-related incidents). Ultimately, this plan ensures all members of the team understand their role and work towards achieving common goals.

Process 3 - monitoring and controlling project Work: this process involves monitoring progress toward meeting defined goals. When variances occur due to delays or unanticipated issues, the project manager develops corrective actions to address these issues. For example, for CRBDAP, the pm will monitor metrics including the number of registered citizens, milestone completion timeliness, and operational status of the system. Software tools such as Microsoft Project or Jira may be utilized to facilitate tracking of progress. As previously mentioned, in Somalia (where unanticipated security-related incidents or equipment availability issues may occur at

any moment), it is essential to establish a mechanism to continuously assess and adjust project performance based upon changing environmental conditions.

3.2 How Scope Management Reduces Project Failure Risks

Scope Management means clearly defining what is included in the project and controlling any changes. This helps the CRBDAP project avoid failure in the following ways:

- It prevents scope creep. Without clear boundaries, government departments may keep adding new requirements, such as adding a vehicle registration module or a tax system. This makes the project bigger, more expensive, and harder to finish. A formal scope plan stops this from happening.
- It helps with accurate planning. When the scope is clear, the Project Manager can estimate the time, cost, and staff needed more accurately. This is very important in Somalia, where the number of trained ICT professionals is limited.
- It aligns all stakeholders. In Somalia, the Federal Government, regional authorities, and international donors all have different expectations. Scope Management ensures all parties agree on what will be delivered, reducing misunderstandings and conflicts.

3.3 Recommended Software Tool: Microsoft Azure DevOps

Microsoft Azure DevOps is a project management tool that helps teams plan, track, and deliver software projects. It is a good choice for CRBDAP because:

- It allows the team to create tasks and features, such as the biometric registration module or the analytics dashboard, and track their progress in one place.
- It integrates with other tools like GitHub for code management and Power BI for reporting, which makes it easier to coordinate all parts of the project.
- It keeps a record of all changes and approvals, which supports transparency and accountability. This is important for Somalia's CRBDAP since the project is funded by international donors who require regular reports.

Question 4: Project Time Management and Value-Based Education

4.1 Three Project Time Management Processes

Process 1- define activities: In this process, you break down your project work into smaller tasks. Examples from CRBDAP include "meeting w/ NIDA to get requirements", "Designing the Citizen Database", "Developing Biometric Registration Feature", and "Testing the System in Banadir". Breaking down your project's work into smaller tasks makes sure everyone understands their responsibilities and avoids confusion. Confusion is one major issue in Somalia due to unclear requirements.

Process 2- develop schedule: In this process, you develop a timeline of when all tasks will begin and end. To develop your timeline, the project manager can use tools including a Gantt Chart and Critical Path Method (CPM) to determine the correct order of tasks. For example, the development schedule should account for Somali-specific issues; i.e., Islamic Holidays such as Ramadan & Eid, Seasonal Weather Issues like the Jilaal Dry Season, etc. Your schedule should allow for extra time in case of any unforeseen problems. Using a realistic schedule allows the team to plan accordingly.

Process 3- control schedule: This process involves monitoring whether the project is meeting its scheduled deadlines. The project manager will monitor progress using a technique called Earned Value Management (EVM). EVM monitors how many hours have been worked by the staff versus what had been planned. When there is an area of lagging behind in the project, the PM may assign additional resources to that part of the project or conduct parts of the project simultaneously to recover lost ground. Regularly reporting progress to stakeholders, e.g., the government and/or donors, builds trust and provides accountability for the CRBDAP project in Somalia.

4.2 Ethical and Moral Responsibilities in the CRBDAP Project

As the Project Manager of CRBDAP for Somalia, there are important ethical duties related to citizen data and team behavior.

Data Privacy Responsibilities: The CRBDAP system will store very sensitive information about millions of Somali citizens, including biometric data, health records, and

home addresses. Many Somalis have experienced conflict and have reasons to be careful about sharing personal information. The Project Manager must make sure that:

- Only necessary data is collected. Collecting extra data that is not needed is not ethical.
- Citizens are informed about what data is collected and how it will be used. This information must be available in Somali language and local dialects so everyone can understand.
- Strong security measures like data encryption and access control are used from the start of the project, not added later.
- The system follows international data protection standards, such as the African Union Malabo Convention, even though Somalia's own laws are still being developed.

Team Professionalism Responsibilities: The Project Manager must also make sure the team works with honesty and respect. This includes:

- Saying no to corruption or bribery, especially in procurement, which is a real challenge in the Somali public sector.
- Treating all team members fairly, regardless of clan background or political connections.
- Respecting intellectual property rights when using third-party software.

4.3 Two Value-Based Practices for Ethical Project Management

Practice 1 – Conduct a Privacy Impact Assessment (PIA): Before collecting any citizen data, the project team should do a Privacy Impact Assessment. This means checking what data will be collected, who will have access to it, and what risks exist. For CRBDAP in Somalia, this is very important because citizen data could be misused in a politically sensitive environment. The PIA results should be reviewed by an independent group that includes community leaders and legal experts from different regions of Somalia. This shows citizens that their data is being handled with care and respect.

Practice 2 – Create a Team Code of Conduct: All project team members should sign a Code of Conduct at the start of the project. This document explains the rules for how team

members should behave, especially regarding data handling, honest communication, and conflict of interest. Regular ethics training sessions, at least every three months, should remind the team of these rules. In the Somali context, linking these ethical values to Islamic principles of Amanah (trustworthiness) and Adl (justice) will make the code more meaningful and culturally relevant to team members.

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